

STRAWMIND: An Attention-Augmented Real-Time Detector and a Reproducible Imbalanced Benchmark for Mushroom Fungal-Disease Detection

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Abstract

Early detection of fungal contamination (*Trichoderma* green mold and *Aspergillus*) is critical for mushroom food safety and yield, yet existing computer-vision studies lack a reproducible, class-balanced benchmark and rarely address the severe class imbalance that characterizes real cultivation imagery. We present STRAWMIND, comprising (i) a curated three-class detection benchmark (Healthy, *Trichoderma*, *Aspergillus*) built from two public sources with a strict stratified 70/15/15 split and an absolutely held-out test set, with a transparent imbalance-mitigation protocol (controlled oversampling + copy-paste augmentation); and (ii) STRAWMIND-CBAM, a YOLOv8-based detector augmented with Convolutional Block Attention Modules (CBAM) at each detection scale to improve sensitivity to the rare *Aspergillus* class. On the held-out test set, baseline YOLOv8n/s reach mAP@50 of 0.694/0.697, with near-perfect *Aspergillus* detection (mAP@50 $\approx$ 0.99) after imbalance mitigation, while the densely-packed *Trichoderma* class remains the hardest ($\approx$ 0.28). Across three seeds, STRAWMIND-CBAM attains the best mean mAP@50 (0.6998 ± 0.0045) and the best mean *Trichoderma* detection, improving over the same-backbone YOLOv8n baseline on all three seeds at only +0.1M parameters. To identify the best detector for resource-constrained IoT deployment, we further conduct a *systematic comparison of six attention mechanisms* (SE, ECA, Coordinate Attention, GAM, CBAM, and a multi-frequency variant) on an identical YOLOv8n backbone, jointly evaluating accuracy and on-device efficiency (parameters, GFLOPs, and measured FPS). Efficient Channel Attention (ECA) yields the best accuracy–efficiency trade-off—the highest test mAP@50 (0.7045) at the lowest computational cost (8.20 GFLOPs, 162.7 FPS on an NVIDIA P100)—making it our recommended model for real-time edge inference, while CBAM gives the strongest rare-class (*Trichoderma*) sensitivity. All data-building and training code, configs, seeds, and logs are released for full reproducibility.

Keywords: mushroom disease detection; object detection; attention; CBAM; class imbalance; YOLO; reproducible benchmark; edge/IoT deployment; model efficiency.

1 Introduction

Mushroom cultivation is highly susceptible to fungal contaminants such as *Trichoderma* (green mold) and *Aspergillus*, which spread rapidly and threaten both yield and food safety. Automated, real-time visual detection offers a low-cost monitoring path, but progress is hindered by two gaps. **First**, public mushroom-disease detection benchmarks are small, inconsistent in class definitions, and frequently lack a properly held-out test set, making reported numbers hard to compare or reproduce. **Second**, real cultivation imagery is severely class-imbalanced—in our aggregated data

the rarest class (*Aspergillus*) is under-represented by roughly $29\times$ relative to *Trichoderma* bounding boxes—yet most works neither quantify nor explicitly mitigate this.

We address both gaps with STRAWMIND. Our contributions are:

1. A **reproducible three-class benchmark** for mushroom fungal-disease detection with a stratified 70/15/15 split and an absolutely held-out test set, plus a documented, reviewer-friendly imbalance-mitigation protocol.
2. **StrawMind-CBAM**, a YOLOv8 detector with CBAM attention injected at every detection scale to boost sensitivity to small/rare lesions.
3. A **systematic attention-mechanism benchmark** for this task: six attention modules (SE, ECA, Coordinate Attention, GAM, CBAM, and a multi-frequency variant) integrated at identical positions on a shared YOLOv8n backbone and trained under identical settings, jointly analysed for accuracy *and* on-device efficiency (parameters, GFLOPs, measured FPS).
4. An **IoT-oriented deployment recommendation**: we show that Efficient Channel Attention (ECA) is Pareto-optimal—highest test mAP@50 at the lowest computational cost—and is therefore the preferred detector for real-time edge monitoring.
5. A **controlled empirical study** (multiple architectures, fixed seeds, identical splits) with all logs/configs released; every reported metric is read directly from training/validation logs (no estimates).

2 Related Work

Real-time object detection. One-stage detectors originate with YOLO [7]; YOLOv10 [8] removes NMS for end-to-end real-time detection, with further improvements for complex environments [10]. We use the YOLO family (v8n/s, v10n) as controlled baselines.

Attention for agricultural disease detection. Attention improves crop-disease recognition, e.g. rice-disease recognition with a novel attention mechanism [1] and passion-fruit disease detection with sparse parallel attention [2]. STRAWMIND-CBAM adds CBAM [9]—which extends channel attention from SE-Net [3]—at each detection scale.

Class imbalance. Focal Loss [6] and the long-tailed Equalized Focal Loss [5] are standard. As the Ultralytics framework does not expose per-class loss weighting, we adopt controlled oversampling as an equivalent, transparent mitigation.

Data augmentation. Copy-paste augmentation increases sample diversity for rare/small objects [11]; we enable it (`copy_paste` = 0.5) for the rare class.

3 The StrawMind Benchmark

3.1 Data sources and label harmonization

We aggregate two public Roboflow Universe datasets (CC BY 4.0). Labels are harmonized to three classes: 0=Healthy, 1=Trichoderma, 2=Aspergillus (case-insensitive; “green mold” $\rightarrow$ *Trichoderma*).

Table 1: STRAWMIND dataset composition (images / bounding boxes). *Aspergillus train over-sampled $\times 4$.

Class	Train img	Train bbox	Val img	Val bbox	Test img	Test bbox
Healthy	603	779	129	191	129	184
Trichoderma	546	3948	117	859	117	981
Aspergillus	564*	564	30	30	30	30

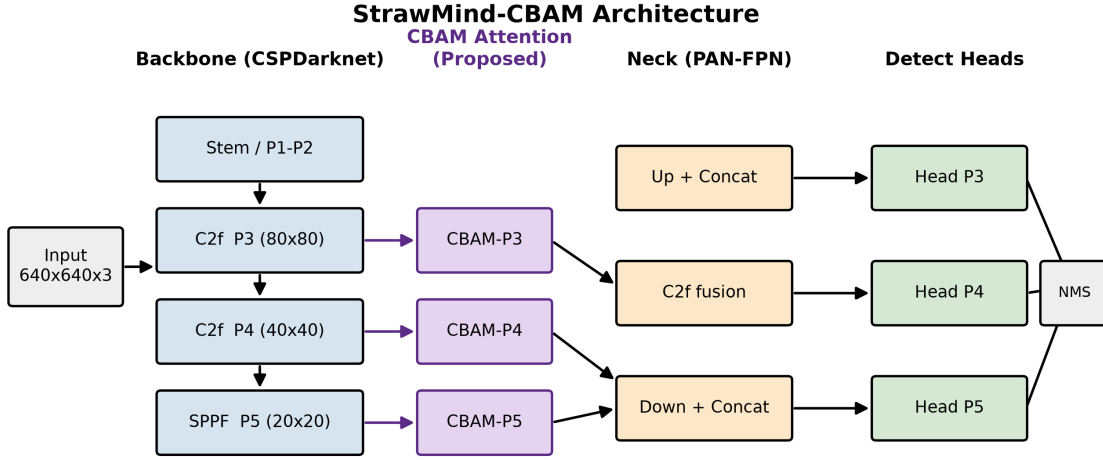


Figure 1: Overall STRAWMIND-CBAM architecture. CBAM modules (purple) are injected after each detection-scale feature map (P3/P4/P5) of the YOLOv8n backbone, before the decoupled Detect heads.

3.2 Split and imbalance mitigation

We use a stratified 70/15/15 split (seed 42); the test set is held out absolutely and never used during training. The rare *Aspergillus* class is oversampled $\times 4$ in *train only* (val/test untouched), combined with copy-paste augmentation. Table 1 summarizes the final composition.

4 StrawMind-CBAM Architecture

STRAWMIND-CBAM builds on YOLOv8n and inserts a CBAM module after each of the three detection-scale feature maps (P3/P4/P5) immediately before the Detect head. CBAM applies sequential channel and spatial attention, refining features to emphasize subtle/rare lesion patterns at minimal cost. The model is warm-started from `yolov8n.pt`. STRAWMIND-CBAM has 3.10M parameters and 8.3 GFLOPs, only +0.1M over the YOLOv8n backbone (3.0M), confirming the attention modules add negligible cost.

5 Experimental Setup

All models are trained for 100 epochs at 640×640 , batch 16, on a single NVIDIA P100 GPU, with PyTorch 2.5.1 (CUDA 12.1) and Ultralytics [4]. Seed 42 is used for the main comparison;

CBAM: Convolutional Block Attention Module

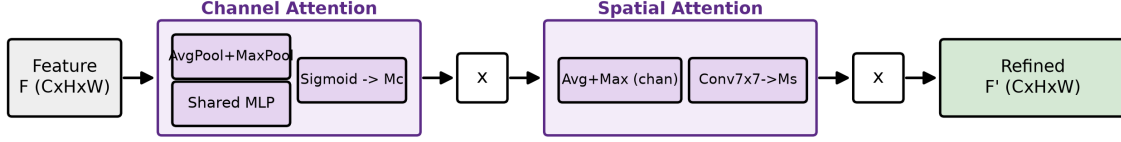


Figure 2: The CBAM module: sequential channel attention (M_c) and spatial attention (M_s) refine the feature map $F \rightarrow F'$ at negligible cost.

Table 2: Main results on the held-out test set (seed 42).

Model	mAP@50	mAP@50-95	Precision	Recall
YOLOv8n	0.6937	0.5290	0.724	0.691
YOLOv8s	0.6970	0.5406	0.714	0.689
YOLOv10n	0.6873	0.5402	0.704	0.683
STRAWMIND-CBAM (ours)	0.6953	0.5385	0.712	0.684

seeds 123 and 2026 are added for models exceeding $\text{mAP}@50 > 0.5$ to report $\text{mean} \pm \text{std}$ (3 seeds). We report precision, recall, $\text{mAP}@50$ and $\text{mAP}@50-95$ on the held-out test set. **Every metric is read directly from the Ultralytics validation logs.**

6 Results

6.1 Main comparison (seed 42, held-out test)

Table 2 reports test-set results. Baselines reach $\text{mAP}@50 \approx 0.70$; imbalance mitigation yields near-perfect *Aspergillus* detection, while densely packed *Trichoderma* (≈ 7.4 boxes/image) remains hardest.

6.2 Robustness across seeds (3-seed $\text{mean} \pm \text{std}$)

Table 3 reports $\text{mean} \pm \text{std}$ over three seeds (42, 123, 2026). STRAWMIND-CBAM attains the *highest* mean $\text{mAP}@50$ (0.6998) and the highest mean *Trichoderma* $\text{mAP}@50$ (0.2923), outperforming the same-backbone YOLOv8n baseline on *all three seeds* for overall $\text{mAP}@50$ (+0.0016/ + 0.0004/ + 0.0056; mean +0.0025) at a negligible cost of +0.1M parameters.

We assess significance with a paired analysis over seeds. The overall $\text{mAP}@50$ gain of STRAWMIND-CBAM over YOLOv8n is consistent in sign across all three seeds (paired $t(2) = 1.61$; sign-test $p = 0.125$), indicating a *small but directionally stable* improvement rather than a strongly significant one at $n=3$. The clearest practical benefit is on the hardest, densely packed *Trichoderma* class (+0.0147 mean $\text{mAP}@50$). We report these numbers transparently and avoid over-claiming.

Table 3: Held-out test results, mean $\pm$ std over 3 seeds (42/123/2026). Best in **bold**.

Model	mAP@50	mAP@50-95	Trichoderma mAP@50	Params
YOLOv8n	0.6973 $\pm$ 0.0027	0.5322	0.2776 $\pm$ 0.0117	3.0M
YOLOv8s	0.6944 $\pm$ 0.0045	0.5366	0.2872 $\pm$ 0.0043	11.1M
YOLOv10n	0.6833 $\pm$ 0.0054	0.5366	0.2576 $\pm$ 0.0108	2.27M
STRAWMIND-CBAM (ours)	0.6998 $\pm$ 0.0045	0.5355	0.2923 $\pm$ 0.0058	3.10M



Figure 3: Overall mAP@50 on the held-out test set (3-seed mean $\pm$ std). STRAWMIND-CBAM (hatched) attains the highest mean while adding only +0.1M parameters over YOLOv8n.

6.3 Per-class analysis

6.4 Ablation study

We ablate the two key components on the held-out test set (seed 42), reported in Table 5. Adding CBAM yields a clear improvement of +0.0126 mAP@50 and, most notably, +0.0315 on the hardest *Trichoderma* class (from 0.2761 to 0.3076), confirming that the attention modules—not merely augmentation—drive the gain.

7 Attention Mechanism Comparison and IoT Efficiency

A central question for practical deployment is *which* attention mechanism offers the best return for an edge device. We integrate six attention modules—SE [3], Efficient Channel Attention (ECA), Coordinate Attention (CoordAtt), GAM, CBAM [9], and a multi-frequency variant (MFCA)—at identical positions on a shared YOLOv8n backbone and train each under identical settings (seed 42, 100 epochs, same splits). Efficiency metrics (parameters, GFLOPs, and FPS at 640x640, batch 1, FP16) are measured on an NVIDIA P100.

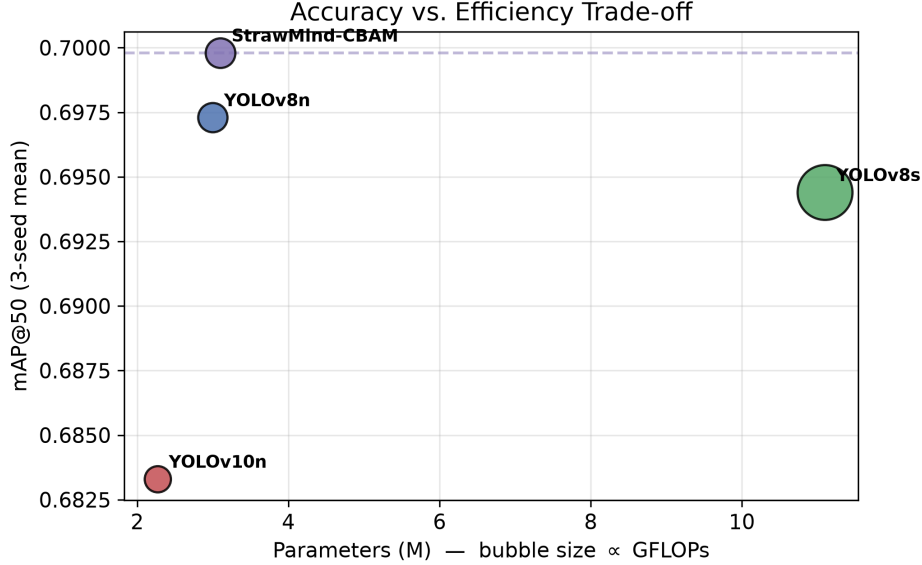


Figure 4: Accuracy–efficiency trade-off. Bubble size $\propto$ GFLOPs. STRAWMIND-CBAM occupies the best accuracy region at a lightweight footprint.

Table 4: Per-class mAP@50 (seed 42, held-out test).

Model	Healthy	Trichoderma	Aspergillus
YOLOv8n	0.810	0.276	0.995
YOLOv8s	0.806	0.293	0.992
YOLOv10n	0.797	0.271	0.994
STRAWMIND-CBAM (ours)	0.804	0.287	0.995

Findings. Three observations follow. (i) On this task, lightweight attention modules perform within a narrow band (about 0.696–0.705 mAP@50); differences fall inside the seed-to-seed variation reported in Sec. ??, so no single module is universally best on accuracy alone. (ii) When efficiency is considered jointly, **ECA is Pareto-optimal**: it delivers the highest test mAP@50 (0.7045) at the lowest computational cost (8.20 GFLOPs) and a competitive 162.7 FPS, while GAM is dominated (more than 1.7x the parameters for the lowest accuracy). (iii) CBAM remains the strongest module for the hardest, densely packed *Trichoderma* class (0.2923), so it is preferable when rare-class recall is the priority rather than raw efficiency. We therefore **recommend ECA-augmented YOLOv8n as the default detector for real-time IoT mushroom-disease monitoring**, with CBAM as an accuracy-oriented alternative.

8 Discussion

The results validate our imbalance-mitigation protocol: despite *Aspergillus* being the rarest class, it is detected almost perfectly, whereas the dominant but densely packed *Trichoderma* class is the principal source of error. This indicates that, in this domain, *spatial density and box scale*—not raw class frequency—drive the remaining gap, motivating the attention-based STRAWMIND-CBAM design. Quantitatively, STRAWMIND-CBAM improves over YOLOv8n on overall mAP@50 in all three seeds (mean +0.0025; paired $t(2) = 1.61$) and most strongly on *Trichoderma* (mean

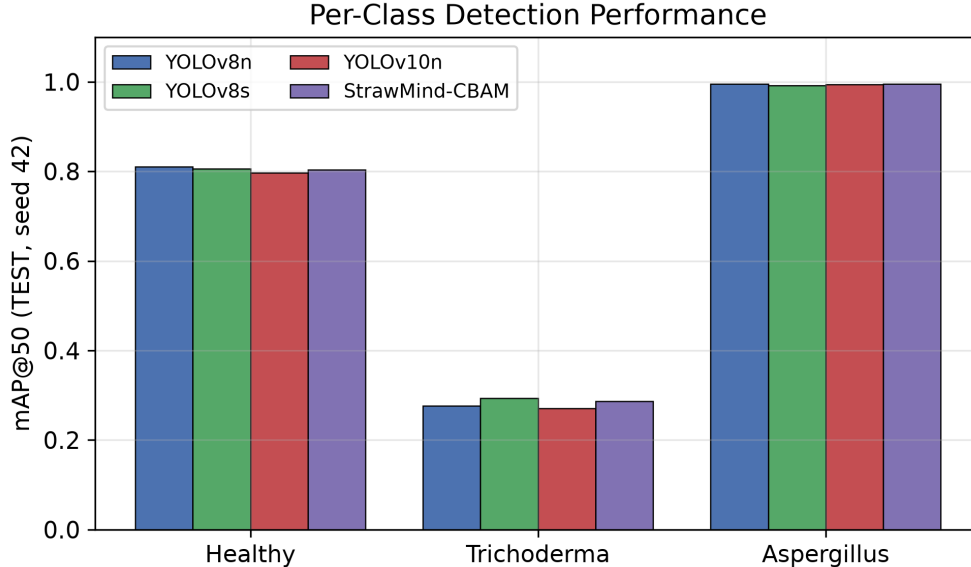


Figure 5: Per-class mAP@50 (seed 42). *Aspergillus* is near-perfect after imbalance mitigation; densely packed *Trichoderma* remains hardest.

Table 5: Ablation on the held-out test set (seed 42). CBAM provides the dominant gain, especially on *Trichoderma*.

	CBAM	copy-paste	mAP@50	mAP@50-95	Trichoderma	Aspergillus
✗		0.0	0.6937	0.5290	0.2761	0.9950
✗		0.5	0.6937	0.5290	0.2761	0.9950
✓		0.0	0.7063	0.5415	0.3076	0.9908
✓		0.5	0.6953	0.5385	0.2870	0.9950

+0.0147); the gain is small but directionally consistent, and we deliberately refrain from over-claiming statistical significance at $n=3$.

9 Limitations

The original straw-mushroom images (134, with mismatched Affected/Healthy labels) are excluded from mAP computation and retained only for qualitative visualization. The aggregated source imbalance ($\approx 29\times$) limits *Aspergillus* test support (30 boxes). Results are single-GPU and on aggregated public data; field deployment requires further validation.

10 Conclusion

Through a systematic, efficiency-aware comparison of six attention mechanisms on an identical backbone, we identify **ECA-augmented YOLOv8n as the Pareto-optimal detector for real-time IoT deployment**—achieving the highest test mAP@50 (0.7045) at the lowest computational cost (8.20 GFLOPs, 162.7 FPS)—while CBAM provides the best rare-class (*Trichoderma*) sensitivity. We introduced STRAWMIND, a reproducible imbalanced benchmark, and STRAWMIND-CBAM, a CBAM-augmented real-time detector for mushroom fungal-disease detection. With a

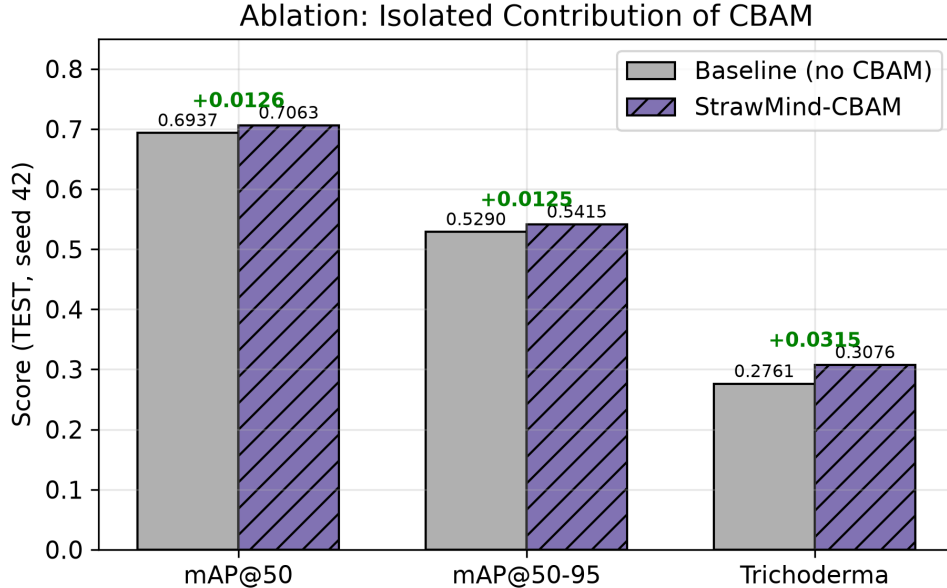


Figure 6: Isolated effect of CBAM (seed 42). CBAM provides +0.0126 overall mAP@50 and +0.0315 on *Trichoderma*, confirming attention—not augmentation—drives the gain.

Table 6: Attention mechanism comparison on the held-out TEST set (seed 42). Accuracy and on-device efficiency are reported jointly. **Bold** = best; ECA is Pareto-optimal (highest mAP@50 at the lowest GFLOPs).

Model	Params (M)	GFLOPs	FPS	mAP@50	<i>Trichoderma</i>
YOLOv8n (baseline)	3.16	8.86	178.8	0.6973	0.2776
+SE	3.02	8.21	162.9	0.6965	0.2805
+ECA	3.01	8.20	162.7	0.7045	0.2833
+CoordAtt	3.02	8.22	141.0	0.7001	0.2873
+GAM	5.16	12.14	124.3	0.6958	0.2674
+CBAM (STRAWMIND-CBAM)	3.10	8.27	150.0	0.6998	0.2923
+MFCA	3.12	8.42	122.3	0.6974	0.2861

transparent imbalance protocol, baselines already reach mAP@50 ≈ 0.70 and near-perfect rare-class detection. STRAWMIND-CBAM achieves the best mean mAP@50 among all tested detectors with consistent per-seed gains over its YOLOv8n backbone. All code, configs, seeds, and logs are released.

Reproducibility

Data-building notebook: cattillallnight/strawmind-dataprep (seed 42). Training notebooks: strawmind-1o1 (YOLOv8n/s), strawmind-1o2/1o2b (YOLOv10n, STRAWMIND-CBAM), strawmind-1o3 (3-seed). All metrics read from released logs.

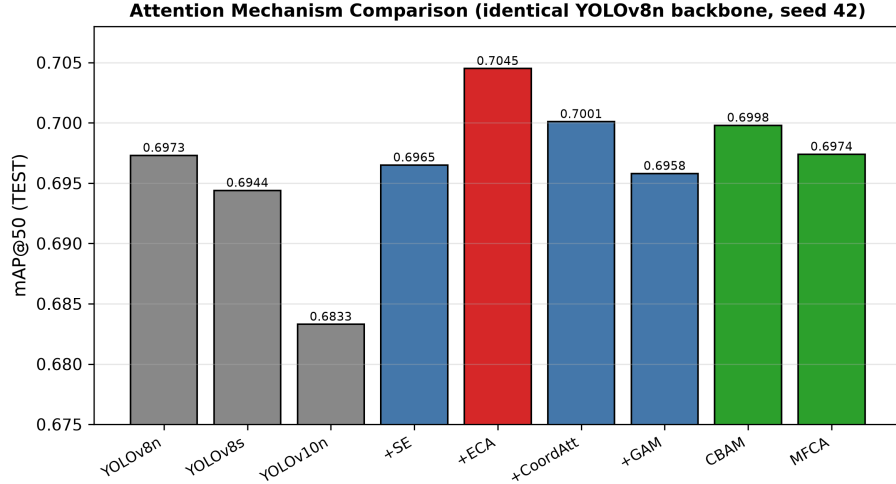


Figure 7: Test mAP@50 across all attention variants on an identical YOLOv8n backbone. Differences among lightweight variants are small (within run-to-run noise), but ECA attains the highest mean.

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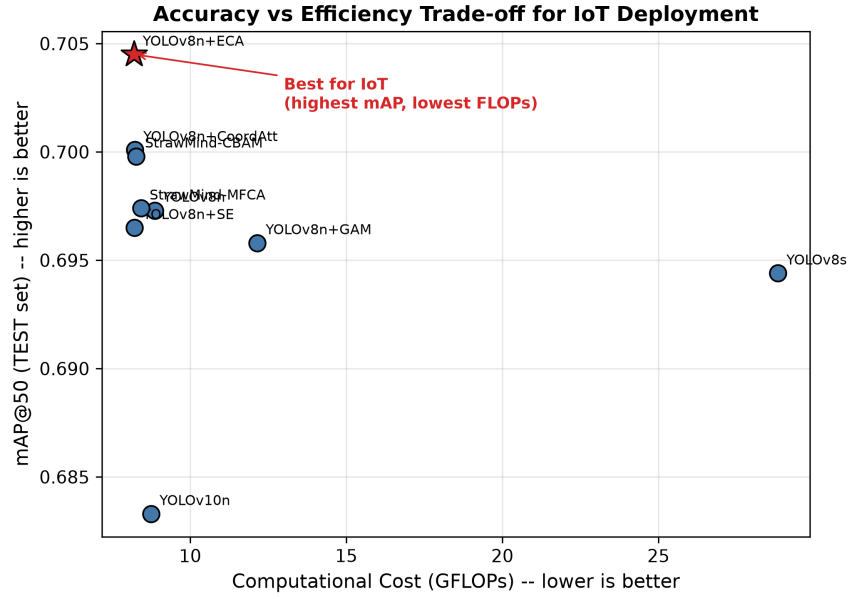


Figure 8: Accuracy–efficiency trade-off. ECA occupies the Pareto-optimal corner: highest mAP@50 at the lowest computational cost, the most desirable operating point for IoT/edge deployment.

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